

Demonstration of Proposed Features

Date	3 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Speech-to-Text Transcription	Converts uploaded audio explanation into text using OpenAI Whisper	Implemented	Yes	Tested with real recorded audio across multiple concepts
2	Semantic Understanding Scoring	Compares transcript meaning to a reference concept using Sentence-BERT	Implemented	Yes	Produces accurate similarity scores even with minor transcription noise
3	Audio Fluency Analysis	Detects filler words, pause ratio, and RMS energy using Librosa	Implemented	Yes	Verified against real speech samples
4	Combined Scoring Engine	Merges semantic and fluency scores into a final weighted score and label	Implemented	Yes	Weighted formula: 60% understanding, 40% fluency
5	AI-Generated Feedback	Gemini API generates a natural-language feedback paragraph based on scores	Implemented	Yes	Includes automatic retry logic for API rate-limit handling
6	PDF Report Generation & Waveform Visualization	Generates a downloadable PDF report with transcript, scores, feedback, and waveform image; also displays the waveform live on the dashboard	Implemented	Yes	Fully integrated into the Streamlit UI

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%